D8 Prompting for correct code

State versions, constraints, expected outputs and tests.

D8-L1 State versions and constraints in every prompt

Every coding prompt names versions, platform and dialect, the constraints from your standard, and what not to do.

```
Environment: Databricks Runtime 17.3 LTS (Spark 4.0, Python 3.12),
PySpark DataFrame API, Delta tables in Unity Catalog.
Local tests: pandas 3.0, pytest 9.
```

Four things every coding prompt states	Versions and platform, the dialect, the relevant standards, and what not to do.
Why name the SQL dialect	Syntax differs by platform. 'SQL' alone invites Snowflake or T-SQL statements.
Where to keep the prompt header	In the repository next to CLAUDE.md, updated when versions change.

D8-L2 State expected outputs and tests in the prompt

Give a worked example, the edge cases with answers, the output format and the tests. Then review against them.

```
Input (respondent_id, reach_flag, final_weight):
  R1, 1, 2.0   R2, 0, 1.0   R3, 1, 1.0   R4, null, 1.0
Expected: weighted_reach = 3.0, weighted_base = 4.0, pct = 0.75
R4 has no reach answer: exclude it from the base.
```

Four things that specify correct code in a prompt	A worked example, edge cases with answers, the output format, and fixed tests.
Instruction that protects your tests	'Make these tests pass without editing them. If a test seems wrong, say so.'
Why work the example out yourself	If you cannot compute the answer by hand, the rule is not clear enough to ask for code.

D8-L3 Iterate on a failing answer

Send the failing test, the error and the smallest input. Fix one thing per round, diff the tests, and stop after two failed rounds.

```
test_zero_share fails:
  assert compare_waves({"a": 0.0}, {"a": 0.12}) == ["a"]
  got [] instead of ["a"]
Line 6 uses `if current_pct and prior_pct:`. I think 0.0 is treated as missing.
Fix only this. Do not change the tests or other behaviour.
```

What to send when AI code fails a test	The failing assertion, actual vs expected, the smallest input, what you checked, and 'fix only this'.
The test-tampering check	Diff the test file after every answer. Removed or changed assertions need your approval.
When to stop iterating	After two focused rounds without progress: debug it yourself, or restart in a fresh conversation.